

# Resumable Online Index Create in SQL Server 2019

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# ABOUT



In professional life consultant, Microsoft Certified Trainer, administrator, developer, freelancer, sometimes a miracle-worker and magician.



From the beginning of career associated with Microsoft technologies.



Private life in numbers: 1 husband, 1 daughter, 1 cat and 2 dogs. My hobbies are detective stories and photography.



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# AGENDA



Intro



Create index



Pause index



Resume index



Create  
another index



Index rebuild



Sort in  
TempDB



Tips

# INTRO

- **SQL Server 2019** provides an important feature **Resumable Online Index Create**.
- In SQL Server 2017, we can only perform a **resumable online index rebuild**.
- We want to create an index on a big table with millions of rows.
- It might take time to create an index on this table.
- While the index creation is in progress, due to some performance issues or failover, we need to cancel the create index statement.
- It can be a big pain again as we have to start it again which will again take time and resources to start the index rebuild all over again.

# INTRO

- In SQL Server 2019, we can **pause** and **resume** the online index creation process similar to resumable online index rebuilds.
- We can **cancel** this process as well without affecting existing indexes.
- Previously we had to stop the index creation, which also takes a lot of time to roll back and then start from the scratch again.

# USAGE CASES

This feature can be useful in the following scenarios occur:

- Database failover
- Disk space issues
- System resource contention
- Minimize transaction log growth

# PREREQUISITES

In order to use this feature, we need to:

- Set database compatibility mode to **SQL Server 2019 (150)**
- Specify **Resumable=ON** for the online index create command
- SQL Server Management Studio (SSMS) **>= 18.0**, which supports the SQL Server 2019 version

## Download SSMS

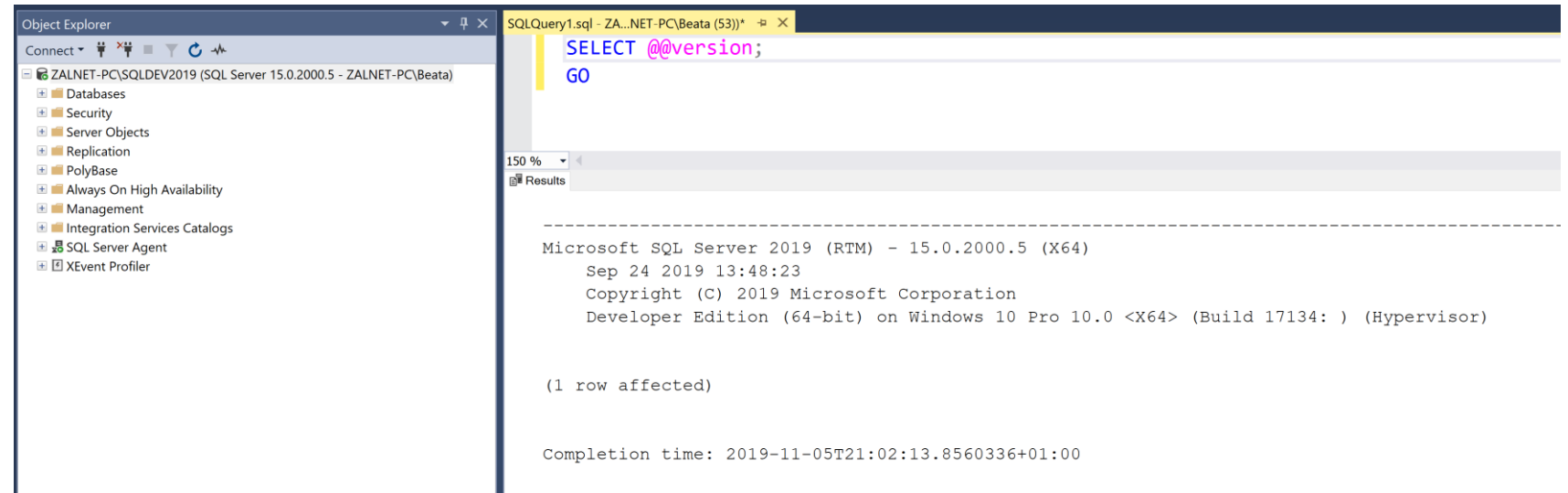
SSMS 18.4 is the latest general availability (GA) version of SSMS. If you have a previous GA version of SSMS 18 installed, installing SSMS 18.4 upgrades it to 18.4. If you have an older *preview* version of SSMS 18.x installed, you must uninstall it before installing SSMS 18.4.

### Version information

- Release number: 18.4
- Build number: 15.0.18206.0
- Release date: November 04, 2019

# SQL Server 2019

- Connect to SQL Server 2019:





# CREATE INDEX

- The **RESUMABLE=ON** parameter in the WITH clause is what makes this CREATE INDEX command resumable.
- Note that you also have to include the **ONLINE = ON** clause when you use the RESUMABLE option.
- If you leave off the ONLINE option, you will get an error.

```
CREATE NONCLUSTERED INDEX CI_tab1_Id  
ON dbo.tab1(NameColumn)  
WITH (ONLINE=ON, RESUMABLE = ON );  
GO
```

## Messages

```
Msg 11438, Level 15, State 1, Line 42  
The RESUMABLE option cannot be set to 'ON' when the ONLINE option is set to 'OFF'.
```

DEMO 1

## Resumable Online Index Create

# ALTER INDEX ... PAUSE

- After allowing the resumable index creation operations to process for a while, you can PAUSE this index create operation, by running the code in a new query window:

```
ALTER INDEX NCI_tab1_NameColumn_AddressColumn ON  
dbo.tab1  
PAUSE;  
GO
```

# CHECK THE INDEX STATE

- Once you have paused your online resumable index operation you can determine how far along the process progressed prior to being paused by using a new system view that came out in the current version of Azure SQL Database, or SQL Server 2017 named [sys.index\\_resumable\\_operations](#).
- By running this code, you can see how far along the online resumable index creation operation progressed prior to being paused.

```
SELECT
    name AS IndexName,
    percent_complete,
    state_desc,
    start_time,
    last_pause_time,
    total_execution_time AS ExecutionMin,
    -- execution time times
    -- ratio of percent to complete and precent completed
    total_execution_time * (100.0-percent_complete)/percent_complete
    AS ApproxExecutionMinLeft
FROM sys.index_resumable_operations;
```

DEMO 2

**ALTER INDEX ... PAUSE**

# ALTER INDEX ...ABORT

```
ALTER INDEX NCI_tab1_NameColumn_AddressColumn ON  
dbo.tab1
```

```
ABORT;
```

```
GO
```

DEMO<sub>3</sub>

**ALTER INDEX ... ABORT**

# ALTER INDEX ... RESUME

- To restart a paused indexing process, all you have to do is run the code in a query window:

```
ALTER INDEX NCI_tab1_NameColumn_AddressColumn ON  
dbo.tab1  
RESUME;  
GO
```



# ALTER INDEX ... RESUME

- When you run the code for resume index, the index creation process will pick up where it left off and continue creating the index.
- You can verify that the index creation process is running again and starts where it left off by querying the view `sys.index_resumable_operations` again.
- The output you get from the second run of this code will show that your index creation process is now running, and the percent completed should be slightly further along than what was shown the first time you ran this code.
- When you restart the index process using the **RESUME** option, the **ALTER INDEX** statement continues to run until the index process completes.
- When the index is done being created you will get a message indicated that the **ALTER INDEX** statement completed successfully in the message window.

DEMO 4

**ALTER INDEX ... RESUME**

# MORE RESUMABLE INDEXES

- We **cannot** run two online resumable index operations on the same table, at the same time, while the original (first) online resumable index process is running.

```
CREATE NONCLUSTERED INDEX NCI_tab1_AddressColumn_2  
ON dbo.tab1(AddressColumn)  
WITH (ONLINE = ON, RESUMABLE = ON);  
GO
```

# ERRORS

- You might be thinking you could start a second non-clustered online resumable online index process on the same table while an existing online reusable index is running.
- However, if you try to start a second non-clustered online resumable index operation, or any index operation, on the same table as one already running you will get the error.

## Messages

```
Msg 10637, Level 16, State 3, Line 165  
Cannot perform this operation on 'object' with  
ID 613577224 as one or more indexes are currently  
in resumable index rebuild state. Please refer to  
sys.index_resumable_operations for more details.
```

## DEMO 5

# CREATE ANOTHER NON-CLUSTERED INDEX

# ALTER INDEX ... REBUILD

- You can also pause and resume an **index rebuild** operation by adding the **RESUMABLE = ON** option to your **ALTER INDEX** statements.

```
ALTER INDEX NCI_tab1_NameColumn_AddressColumn ON dbo.tab1 REBUILD  
WITH (ONLINE = ON, RESUMABLE = ON);  
GO
```

## DEMO 6

## ALTER INDEX ... REBUILD

# CREATE INDEX ... SORT IN TEMPDB

- We cannot create a resumable index in tempdb.

```
CREATE NONCLUSTERED INDEX NCI_tab1_NameColumn  
ON dbo.tab1(NameColumn)  
WITH (ONLINE=ON, RESUMABLE = ON, SORT_IN_TEMPDB = ON);  
GO
```

## Messages

```
Msg 11438, Level 15, State 2, Line 207  
The SORT_IN_TEMPDB option cannot be set to 'ON'  
when the RESUMABLE option is set to 'ON'.
```



DEMO 7

**CREATE INDEX ... SORT IN  
TEMPDB**

# TIPS

- You can only pause the index creation process if you have specified the **RESUMABLE** option on the create or rebuild index statement.
- If you already have scripts to create or rebuild your indexes online don't forget to put this new option in your scripts if you want to pause and resume your indexes.
- Also note you can only use the **RESUMABLE=ON** option if you also include the **ONLINE=ON** option as well.

# TIPS

- When you pause your resumable create index or rebuild operation, the session running the index create or rebuild statement will be cancelled and disconnected.
- The messages you get when this happens seem is a little different than I would have expected.
- It would be nice if the messages would have said something like : **Your reusable index operation has been PAUSED.**
- To restart use the **ALTER INDEX** with the **RESUME** option.
- The message about being disconnected is the message that was used to support the online resumable **rebuild** operation in the current Azure SQL database, or SQL Server 2017.

# TIPS

- You **cannot cancel** your resumable index create or rebuild operation by just terminating these index operations.
- When you kill or cancel your resumable index operations the index goes into a **PAUSED** state.
- To truly cancel your resumable index operation completely you will need run an **ALTER INDEX** statement with the **ABORT** option.
- You can only create one resumable online index operation on a table at a time.

# TIPS

- When using the **RESUMABLE** option you can also add a **MAX\_DURATION** option.
- By using the **MAX\_DURATION**, you can set the number of minutes a resumable index create or rebuild operation will run.
- When that time limit is reached, the resumable index process will be killed, in affect pausing the index creation or rebuild operation.
- This would be a useful option to use when you want to perform your index operation in a piecemeal approach.
- When creating a new index using the **RESUMABLE** option, you will not see it listed in **sys.indexes** or in SSMS until the operation has completed.
- To find the index, use the **sys.index\_resumable\_operations** system view.

# Q & A

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Demo:

<https://github.com/bzalewa/DataCommunityLublinNovember2019>

Thank you for your precious time 😊